

SOC Threat Detection & Log Analyzer

Product Requirements Document (PRD)

Version: 1.0

1. Executive Summary

The SOC Threat Detection & Log Analyzer is a cybersecurity analytics platform designed to simulate Security Operations Center (SOC) workflows. The platform enables users to upload security logs, detect suspicious activities, classify security risks, investigate incidents, and generate actionable reports.

The solution combines threat detection, anomaly analysis, attack simulation, and visualization tools to provide an end-to-end security monitoring experience.

2. Problem Statement

Organizations generate large volumes of log data from servers, applications, and network devices. Manual review of these logs is time-consuming and often ineffective.

Common challenges include:

- Difficulty identifying threats within large datasets
- Delayed incident detection
- Limited visibility into attack patterns
- Lack of structured security reporting
- Insufficient training environments for cybersecurity learners

The SOC Threat Detection & Log Analyzer addresses these challenges by automating log analysis and threat identification.

3. Product Vision

To provide an intelligent and accessible SOC simulation platform that empowers security analysts, students, and researchers to identify threats, investigate incidents, and improve cybersecurity awareness through practical log analysis.

4. Product Goals

Primary Goals

- Automate security log analysis
- Detect common cyberattack patterns
- Classify security risks automatically
- Provide actionable threat intelligence
- Visualize attack activity effectively

Business Goals

- Improve threat detection efficiency
- Reduce incident investigation time
- Support cybersecurity education and training
- Demonstrate real-world SOC workflows

5. Stakeholders

Primary Stakeholders

- Security Analysts
- SOC Engineers
- Cybersecurity Students
- Security Researchers

Secondary Stakeholders

- Academic Institutions
- Cybersecurity Training Organizations
- Security Teams

6. User Personas

Security Analyst

Needs

- Automated threat detection
- Incident prioritization
- Security reporting

Pain Points

- Alert fatigue
- Large log volumes
- Limited investigation time

Cybersecurity Student

Needs

- Practical SOC experience
- Exposure to attack detection techniques
- Hands-on learning environment

Pain Points

- Limited access to real SOC tools
- Lack of realistic security datasets

7. Functional Requirements

FR-01 Log Upload

The system shall allow users to upload log files for analysis.

Acceptance Criteria

- Support standard log file formats.
- Successfully ingest and parse uploaded logs.

FR-02 Log Parsing

The system shall convert raw log data into structured security events.

Acceptance Criteria

- Extract timestamps, IP addresses, usernames, and event types.

FR-03 Brute Force Detection

The system shall identify repeated authentication failures.

Acceptance Criteria

- Alert generated when predefined thresholds are exceeded.

FR-04 Suspicious Login Detection

The system shall identify abnormal login activity.

Acceptance Criteria

- Login anomalies detected and reported.

FR-05 Multi-IP Attack Detection

The system shall identify coordinated attack attempts originating from multiple IP addresses.

Acceptance Criteria

- Associated attack events grouped and flagged.

FR-06 Time-Based Anomaly Detection

The system shall identify suspicious activity occurring outside expected operating hours.

Acceptance Criteria

- Events categorized as anomalous when timing thresholds are exceeded.

FR-07 Geo-IP Anomaly Detection

The system shall identify logins from unusual geographical locations.

Acceptance Criteria

- Geographic anomalies highlighted in reports.

FR-08 Risk Assessment Engine

The system shall classify detected events according to risk severity.

Risk Levels

- Low
- Medium
- High

Acceptance Criteria

- Severity assigned automatically.

FR-09 Threat Intelligence Module

The system shall enrich suspicious IP addresses with threat intelligence information.

Acceptance Criteria

- Known malicious IPs flagged appropriately.

FR-10 Dashboard Analytics

The system shall provide interactive security visualizations.

Analytics Components

- Threat Overview
- Alert Dashboard
- Attack Timeline
- Top Attacking IPs
- Attack Frequency Heatmap
- User Activity Tracking

Acceptance Criteria

- Visualizations update dynamically based on analysis results.

FR-11 Attack Simulator

The system shall generate synthetic attack logs for testing.

Supported Scenarios

- Brute Force
- Multi-IP Attack
- Geo-IP Anomaly
- Time Anomaly
- Combined Attack Simulation

Acceptance Criteria

- Generated logs compatible with detection engine.

FR-12 Incident Reporting

The system shall generate downloadable security reports.

Supported Formats

- TXT
- PDF

Acceptance Criteria

- Reports summarize findings accurately.

8. Non-Functional Requirements

Performance

- Log processing completed within 10 seconds for standard datasets.
- Dashboard response time below 3 seconds.

Scalability

- Support increasing log volume through modular architecture.

Reliability

- Consistent threat detection results.
- Stable operation during extended usage.

Security

- Secure handling of uploaded files.
- Validation of user input.
- Protection against malicious uploads.

Usability

- Intuitive dashboard design.
- Minimal learning curve.
- Responsive interface.

Maintainability

- Modular detection architecture.
- Independent security modules.
- Easy addition of new detection rules.

9. System Architecture

Presentation Layer

- Streamlit Dashboard

Business Logic Layer

- Detection Engine
- Risk Engine
- Threat Intelligence Module
- Attack Simulator
- Reporting Engine

Data Layer

- Uploaded Logs
- Processed Security Events
- Generated Reports

10. Risks

Risk	Impact	Mitigation
False Positives	Medium	Rule tuning and threshold optimization
False Negatives	High	Continuous rule improvement
Large Log Files	Medium	Efficient parsing and filtering
Malicious File Uploads	High	File validation and sanitization

11. Product Roadmap

Phase 1 (Current Release)

- Log Upload
- Threat Detection
- Risk Classification
- Dashboard Analytics
- Attack Simulator
- PDF Reporting

Phase 2

- Real-Time Monitoring
- Threat Intelligence APIs
- Role-Based Authentication
- Incident Workflow Tracking

Phase 3

- Machine Learning Detection
- Behavioral Analytics
- SIEM Integration
- Cloud Deployment Scaling
- MITRE ATT&CK Framework Mapping

12. Success Metrics

Operational Metrics

- Detection Accuracy
- Alert Generation Time
- Report Generation Time

User Metrics

- Number of Logs Analyzed
- Dashboard Usage Frequency
- Attack Simulations Executed

Security Metrics

- Threats Detected
- High-Risk Events Identified
- Incident Investigation Efficiency

Conclusion

The SOC Threat Detection & Log Analyzer provides a practical and scalable cybersecurity platform for threat detection, security monitoring, and SOC workflow simulation. The product combines automated analysis, threat intelligence, and visualization capabilities to deliver meaningful security insights while serving as an effective learning and operational tool.